

Warm-up

For each rational function, write the form of the partial fraction decomposition:

$$(1) \int \frac{x^2 + x - 4}{(x+3)^2(x+1)} dx, \quad (2) \int \frac{4x^2 - x + 10}{x^3 - 2x^2 + 4x - 8} dx, \quad (3) \int \frac{1}{(x^2 - 3x + 2)(x^2 + 2x + 2)^2} dx.$$

Medium

(1) Compute $\int \frac{3x}{(x+4)(x-2)} dx$.

(2) Compute the integral (1) from the Warm-up section.

(3) Compute the integral (2) from the Warm-up section.

(4) Compute $\int \frac{x^3}{x^2 - 3x + 2} dx$.

(5) Compute $\int \frac{5}{(x-1)(x^2 + 2x + 2)} dx$.

Stretch

(1) Possibly in the course of finding a partial fractions decomposition, you may come across a term with the square of an irreducible quadratic in the denominator and something linear in the numerator, for example in Warm-up (3). Integrate the following example of such a term, using a trig substitution where appropriate:

$$\int \frac{x+1}{(x^2+1)^2} dx.$$

(2) Compute the same integral using the following *reduction formula*:

$$\int \frac{1}{(x^2+1)^n} dx = \frac{1}{2n-2} \cdot \frac{x}{(x^2+1)^{n-1}} + \frac{2n-3}{2n-2} \int \frac{1}{(x^2+1)^{n-1}} dx.$$

Extra bonus: use integration by parts to derive this formula.